BASAVA HARSHA

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SUMMARY

Motivated, Enthusiastic and dedicated Computer Science Engineering student specializing in Artificial Intelligence and Machine Learning. Proficient in Python and Java with a solid understanding of programming fundamentals and problem-solving techniques. Demonstrates strong analytical thinking, adaptability, and a commitment to continuous learning. Eager to apply technical knowledge and AI concepts to develop innovative solutions for real-world challenges.

EDUCATION

Bachelor of Technology in Computer Science & Engineering
Kalasalingam Academy of Research & Education, Tamil Nadu (**Grade Point: 8.71/10**) — **Expected** 2027

Intermediate Education
Sri Chaitanya Mahila Jr Kalasala, Andhra Pradesh **(Grade Point: 10/10)** — 2023

Board of Secondary Education
Sri Chaithanya High School, Andhra Pradesh **(Grade Point: 10/10)** — 2021

SKILLS

- Programming Languages : Python, Java
- Data Analysis & Visualization Tools : Microsoft Tools, Power Bi
- Artificial intelligence : Prompt Engineering, Responsible Ai, Vibe Coding
- Software Tools : VM VirtualBox, Docker
- Soft Skills : Problem solving, Time Management, Adaptability

PROJECTS

1. **Adaptive Explainable AI for Cyber Threat Detection** | Python, FastAPI, Streamlit, PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, SHAP, LIME, SQLAlchemy, Plotly

   Developed a production-grade AI-powered cybersecurity platform capable of detecting phishing emails, malicious URLs, suspicious login behavior, and network anomalies using Machine Learning, Deep Learning, and Explainable AI techniques. Engineered a modular multi-model threat detection pipeline integrating DistilBERT/BERT for phishing detection, XGBoost for malicious URL and network intrusion detection, and Isolation Forest for behavioral anomaly detection, with an Adaptive Threat Fusion Engine generating unified risk scores and actionable security insights. Implemented Explainable AI using SHAP and LIME to provide transparent model predictions, feature-level explanations, confidence scores, and remediation recommendations while exposing platform functionality through scalable FastAPI REST APIs and an interactive Streamlit Security Operations Center (SOC) dashboard featuring real-time analytics, threat timelines, system monitoring, PDF/CSV report generation, and simulation mode. Designed a production-ready software architecture following SOLID principles with automated testing, CI/CD pipelines, Docker support, GitHub Actions, SQLite/PostgreSQL integration, and cloud deployment on Render and Streamlit Community Cloud, ensuring a scalable, secure, and enterprise-ready cybersecurity platform suitable for academic research and real-world security monitoring.

2. **Custom AI Agent with Memory** | Python, FastAPI, LangGraph, LangChain, Streamlit, Groq API

   Developed a production-ready AI assistant with persistent long-term memory supporting both voice and text interactions, enabling personalized conversations by remembering user preferences, facts, and goals across multiple sessions. Engineered a LangGraph ReAct agent integrated with LangChain and Groq's Llama 3.3 70B model, implementing intelligent memory retrieval, automated memory storage, and contextual response generation using scoped user memories. Implemented a FastAPI backend exposing REST APIs for chat, memory management, and Whisper-based speech transcription while developing a Streamlit frontend featuring voice input, real-time conversations, and an interactive memory dashboard. Designed a scalable multi-user architecture with isolated memory stores, Dockerized deployment, and cloud hosting on Render and Streamlit Community Cloud, delivering a secure, scalable, and production-ready AI assistant supporting persistent conversations and high-volume daily requests.

3. **AI Code Review Bot – GitHub PR Auto-Reviewer** | Python, FastAPI, GitHub Webhooks, Groq API

   Developed an AI-powered GitHub Pull Request review automation system that performs intelligent code analysis and posts detailed review comments directly on GitHub using Groq's Llama 3.3 70B model. Engineered a FastAPI backend integrated with GitHub Webhooks and GitHub REST APIs to automatically retrieve pull request diffs, analyze code quality, and generate actionable feedback for every submitted pull request. Implemented automated detection of logic errors, security vulnerabilities, performance bottlenecks, coding standard violations, and best practice recommendations while providing file-level review comments and overall approval or rejection verdicts. Deployed the backend on Render using zero-cost cloud infrastructure, ensuring a scalable, reliable, and production-ready AI-assisted code review platform without relying on paid AI services.

4. **TravelWear AI – Smart Weather Based Outfit Planner** | React.js, Node.js, MongoDB

   Developed a full-stack AI-powered travel fashion assistant that recommends clothing based on real-time weather conditions retrieved from the OpenWeatherMap API for destinations worldwide. Implemented a Smart Style Advisor using Canvas API color histogram analysis to detect users' skin tones from uploaded images and generate personalized outfit color recommendations by combining skin tone analysis with destination weather conditions. Built an AI chatbot providing destination-specific clothing suggestions, food recommendations, and places-to-visit guidance, along with a smart packing checklist generator featuring PDF export, a 5-day weather forecast with travel date selection, and JWT-based user authentication with user data stored in MongoDB Atlas. Deployed the frontend on Vercel and the backend on Render using environment-based configuration for scalable and reliable production deployment.

CERTIFICATIONS

Introduction to Generative AI – Google Cloud
AI Fundamentals – IBM
Introduction to AI in Azure – Microsoft
Machine Learning Fundamentals – AWS